

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458661.8464 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.046
Transit depth (Rp/Rs)^2	1.94 +/- 1.27 [%]
Orbital Inclination (inc)	83.23 +/- 2.89
Airmass coefficient 1 (a1)	0.78 +/- 0.2
Airmass coefficient 2 (a2)	0.18 +/- 0.17
Scatter in the residuals of the lightcurve fit is	24.48 %
Best Comparison Star	#1 - [314, 397]
Optimal Aperture	11.49
Optimal Annulus	13.96
Transit Duration (day)	0.047 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.046 vs 0.1326 (deviation 0.13σ)
Duration fit vs published	0.047 d vs 0.0737 d (deviation 0.68σ)
Mid-time O–C	13.4 min
Depth SNR	2.8
Residual scatter	24.48 %

Light curve

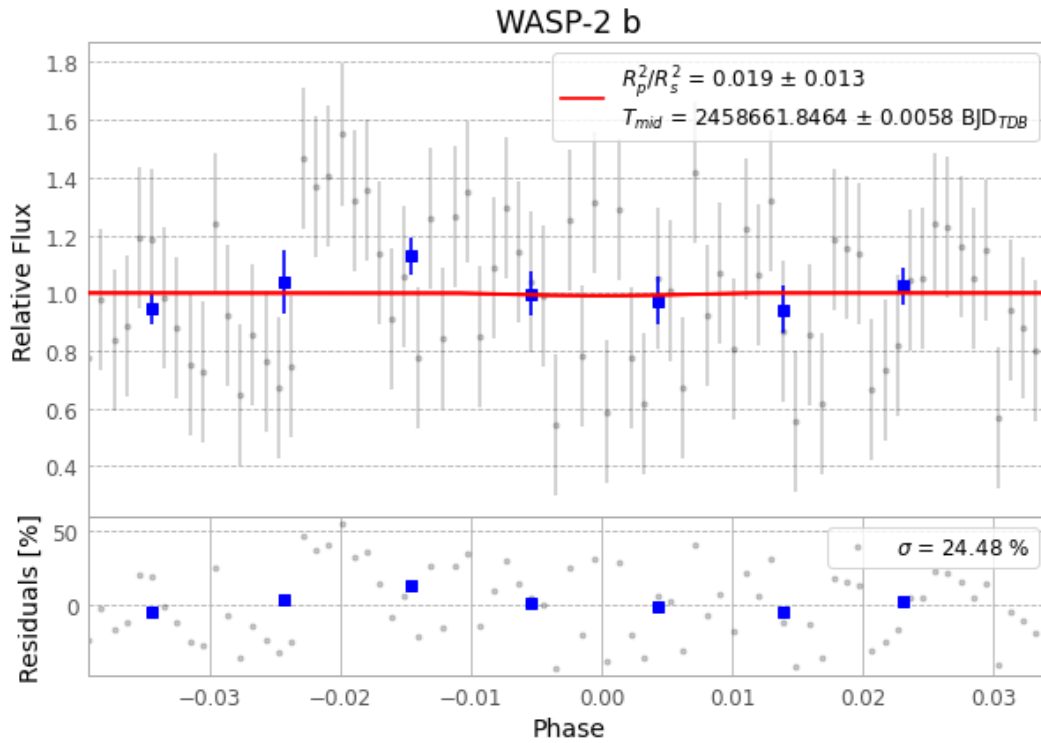


Figure 1 — FinalLightCurve_WASP-2 b_2019-06-26.png (SHA-256 97446b7c9cf3d936...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 394], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 31786ef735791551...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-2190627060912.FITS → WASP-2190627095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-190627.log (a78934f194cd...), FinalLightCurve_WASP-2 b_2019-06-26.png (97446b7c9cf3...), FinalLightCurve_WASP-2 b_2019-06-26.csv (aa56d0f6f4f6...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:37Z.